



Email spam Detection

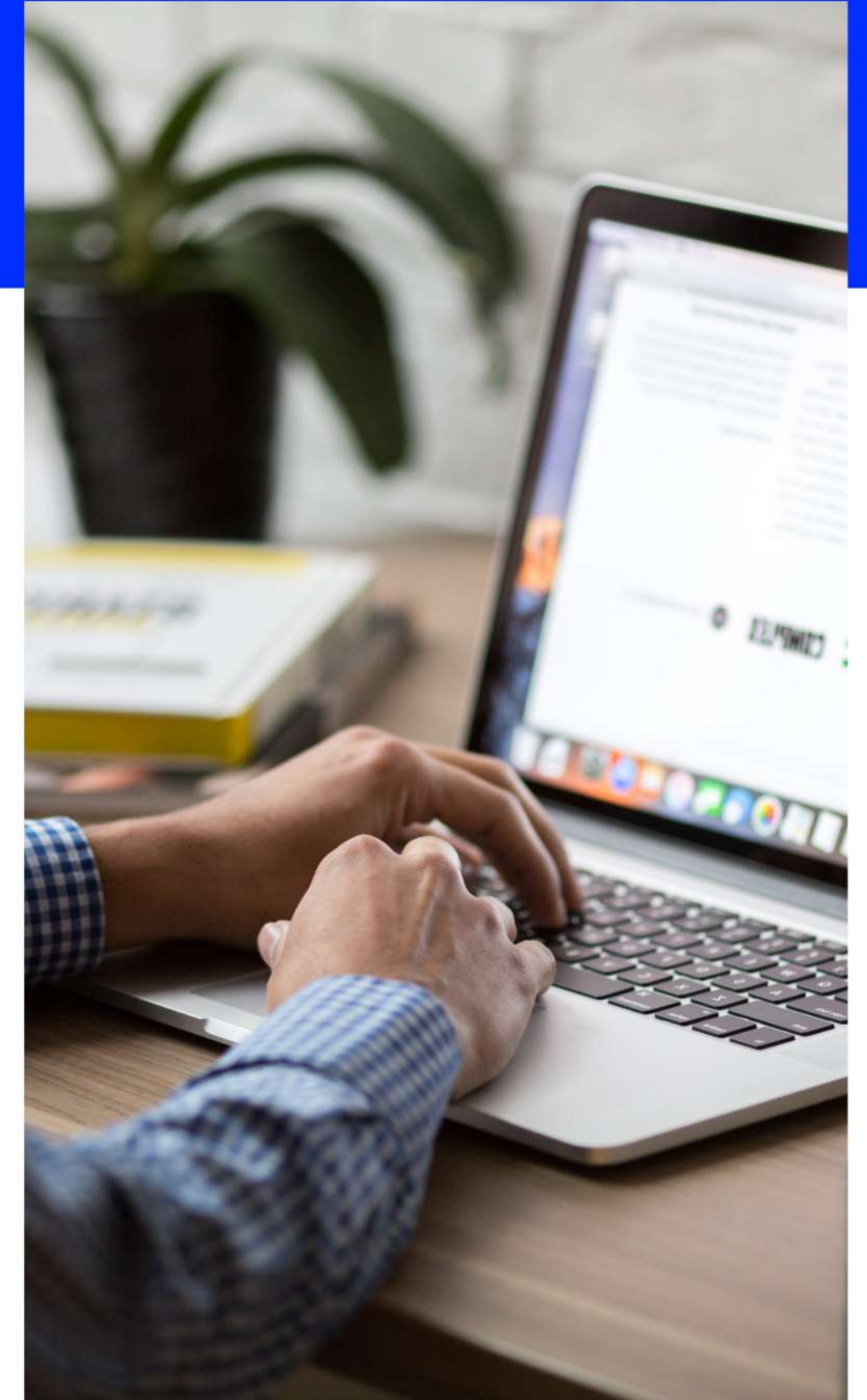
Detect email spam using multiple classification algorithms

Email has become one of the most widely used forms of communication in today's digital world. However, the increasing number of spam messages—such as unwanted advertisements, phishing emails, and fraudulent content—has created serious challenges for users.

Manual filtering of spam emails is time-consuming and inefficient. To overcome this issue, machine learning techniques can be used to automatically detect and classify emails as spam or not spam.

This project focuses on developing an Email Spam Detection system using multiple classification algorithms such as Logistic Regression, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN). The system processes and analyzes email text, converts it into numerical form using TF-IDF, and applies these algorithms to accurately classify messages.

Introduction



Problem statement

Email spam detection in ML



In today's digital world, email communication is widely used, but it is also heavily affected by spam messages, which can be annoying, misleading, or even harmful. Manually filtering spam emails is time-consuming and inefficient.

The objective of this project is to develop a machine learning-based system that can automatically classify emails as spam or not spam (ham) using text analysis techniques. The system will utilize multiple classification algorithms such as Logistic Regression, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN) to analyze and compare their performance.

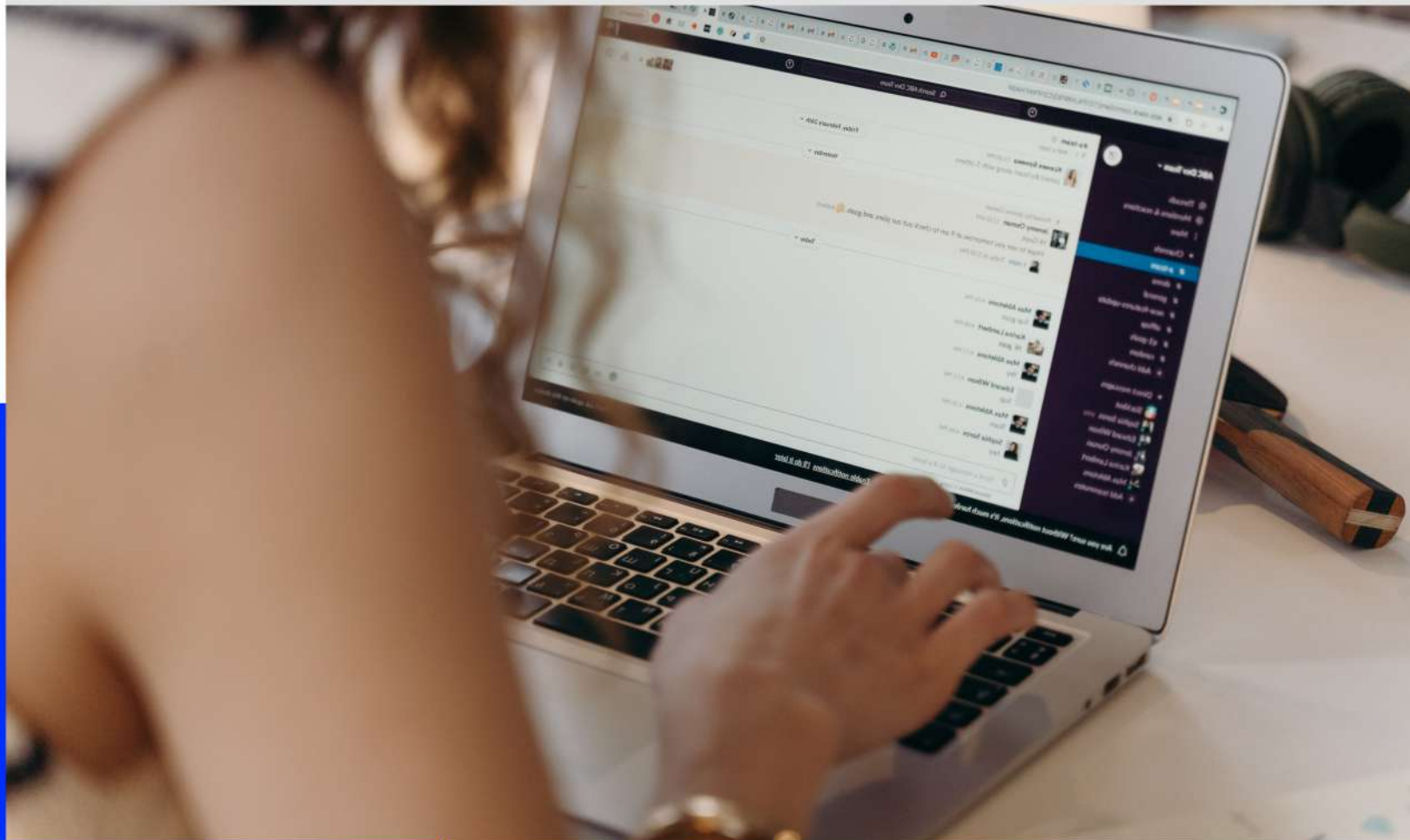
The project involves preprocessing textual data, converting it into numerical form using techniques like TF-IDF, training different models, and evaluating them based on performance metrics such as accuracy, precision, recall, and F1-score.

The goal is to identify the most efficient algorithm for spam detection and build a reliable model that can be used for real-world email filtering.



1. This project aims to build a machine learning system to classify emails as spam or not spam (ham).
2. It helps in automatically filtering unwanted messages, improving user experience and security.
3. The dataset used contains labeled messages (spam and ham).
4. Data preprocessing is performed to clean the text:
5. Convert text to lowercase
6. Remove special characters
7. Remove stopwords
8. The cleaned text is converted into numerical form using TF-IDF, which highlights important words.
9. Three algorithms are used:
10. Logistic Regression (simple and fast)
11. Support Vector Machine (SVM) (high accuracy)
12. K-Nearest Neighbors (KNN) (based on similarity)
13. The dataset is split into training and testing data to evaluate performance.

Explanation of the project



Models are evaluated using:

Accuracy

Precision

Recall

F1-score

Among all models, SVM gives the best performance.
The final system can predict whether a new message is spam or not.





Key Points:

Simple and fast
Works well for linearly separable data
Easy to interpret



In this project, multiple machine learning classification algorithms are used to identify whether an email is spam or not. Each algorithm follows a different approach to learn patterns from the data and make predictions.

Using multiple algorithms helps in comparing their performance and selecting the most accurate and efficient model for spam detection. These algorithms are trained on preprocessed text data and evaluated using standard performance metrics.

The following are the machine learning algorithms used in this project:

1. Logistic Regression

A supervised learning algorithm used for binary classification

Predicts probability between 0 and 1 using a sigmoid function

Classifies data into two classes (e.g., spam / not spam)

Description of the model



2. Support Vector Machine (SVM)

A supervised learning algorithm used for classification

Finds the optimal hyperplane that separates data into classes

Key Points:

Works well with high-dimensional data (like text)

Uses kernel functions (linear, polynomial, RBF)

Provides high accuracy

3. K-Nearest Neighbors (KNN)

A lazy learning algorithm (no training phase)

Classifies data based on nearest neighbors

Key Points:

Uses distance (Euclidean)

Simple to understand

Slow for large datasets

4. Perceptron

A basic neural network model (single-layer)

Used for binary classification

Key Points:

Uses weighted sum + activation function

Learns by adjusting weights

Works only for linearly separable data

5. Decision Tree Classifier

A tree-based supervised learning algorithm

Splits data into branches based on conditions

Key Points:

Easy to understand and visualize

Handles both numerical and categorical data

Can lead to overfitting if not controlled

6. K-Neighbors Classifier

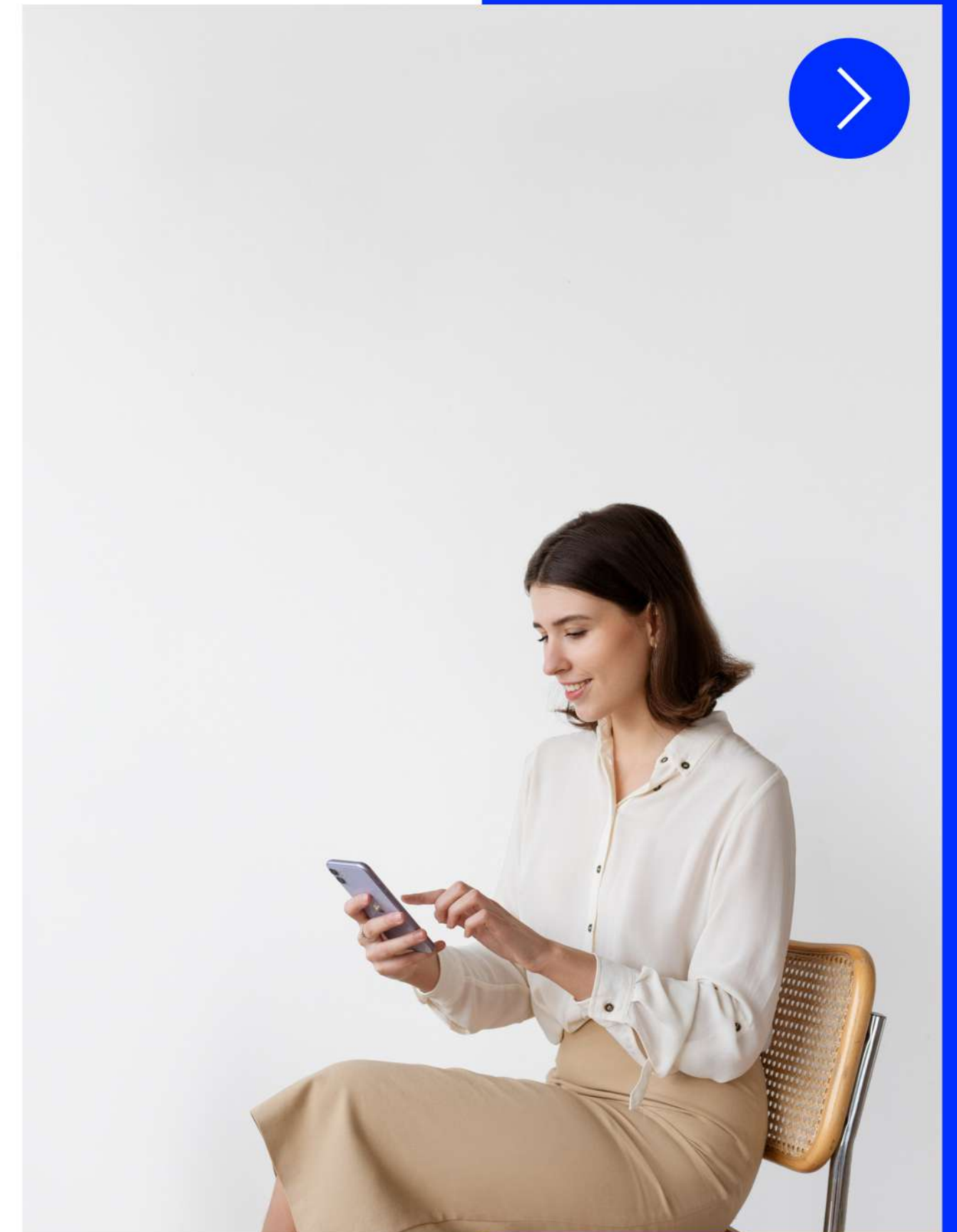
Same as K-Nearest Neighbors (KNN) (just different naming in libraries like sklearn)

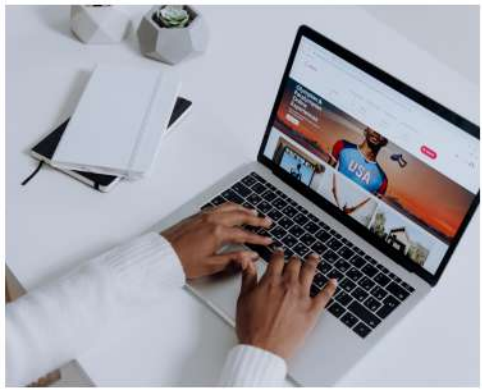
Key Points:

Classifies based on majority class of nearest neighbors

Uses distance metrics

No explicit training phase





performance Metrics of the model

1. Accuracy

Measures the overall correctness of the model
It is the ratio of correct predictions to total predictions

Formula:

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

Meaning:

Higher accuracy → better model performance

2. Precision

Measures how many predicted spam messages are actually spam

Formula:

$$\text{Precision} = TP / (TP + FP)$$

Meaning:

High precision → fewer false alarms (important for spam detection)





3 F1-Score

Harmonic mean of precision and recall

Formula:

$$F1\text{-score} = 2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall})$$

Meaning:

Balances precision and recall

4. Confusion Matrix

A table used to visualize model performance

Predicted Spam Predicted Ham

Actual Spam TP FN

Actual Ham FP TN

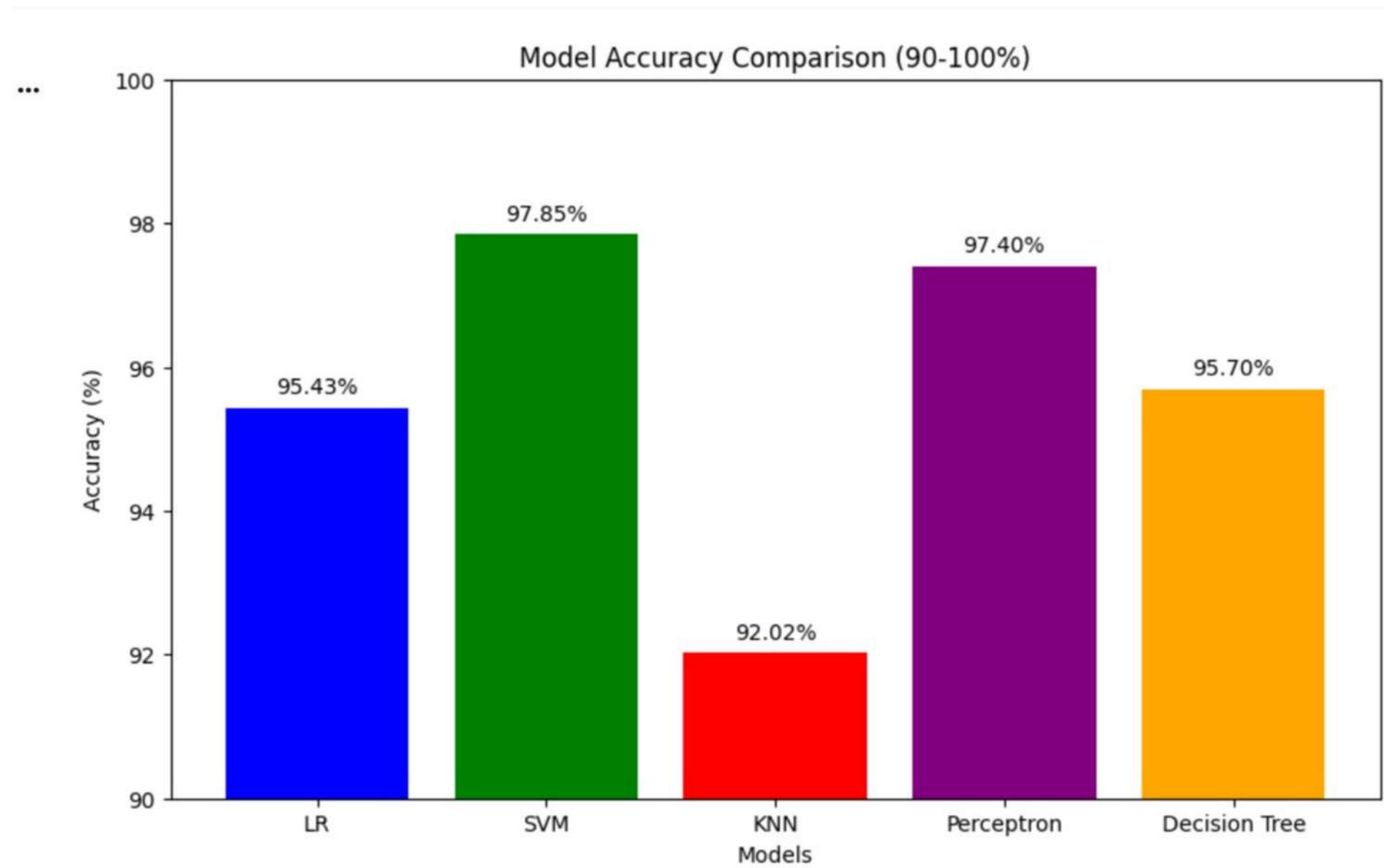
Typical Results (Your Project)

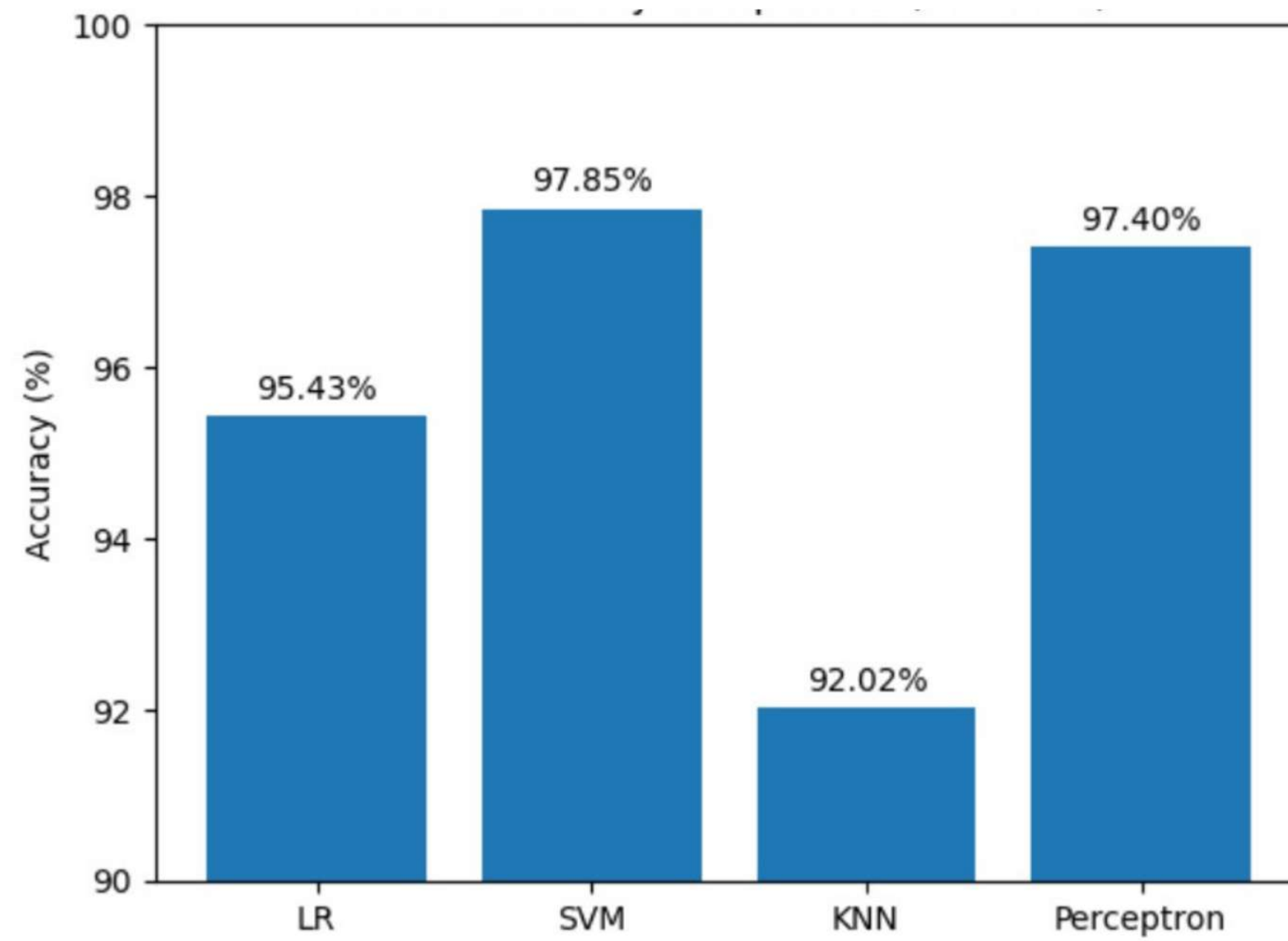
Logistic Regression → ~95% Accuracy

SVM → ~97–99% Accuracy (Best)

KNN → ~90–95% Accuracy

outputs





Thank You

This project successfully demonstrates the use of machine learning techniques to automatically classify emails as spam or not spam. By applying data preprocessing and TF-IDF feature extraction, the text data was effectively converted into a format suitable for model training.

<https://github.com/2403A51186/EMAIL-SPAM-DETECTION->

